The Ramanujan Journal (2025) 68:26
https://doi.org/10.1007/s11139-025-01184-6

Counting imaginary quadratic fields with an ideal class group of 5-rank at least 2

Kollin Bartz[1] · Aaron Levin[1] · Aman Dhruva Thamminana[1]

Received: 31 January 2025 / Accepted: 18 July 2025 / Published online: 1 August 2025

Abstract

We prove that there are $\gg \frac{X^{\frac{1}{3}}}{(\log X)^2}$ imaginary quadratic fields k with discriminant $|d_k| \leq X$ and an ideal class group of 5-rank at least 2. This improves a result of Byeon, who proved the lower bound $\gg X^{\frac{1}{4}}$ in the same setting. We use a method of Howe, Leprévost, and Poonen to construct a genus 2 curve C over $\mathbb{Q}$ such that C has a rational Weierstrass point and the Jacobian of C has a rational torsion subgroup of 5-rank 2. We deduce the main result from the existence of the curve C and a quantitative result of Kulkarni and the second author.

Keywords Ideal class groups · Torsion in Jacobians · 5-rank · Imaginary quadratic fields

Mathematics Subject Classification Primary: 11R29 · Secondary: 11G30 · 14H40

1 Introduction

Given an integer $m > 1$, it has been known since Nagell's 1922 result [14] that there are infinitely many imaginary quadratic number fields with class number divisible by m; the analogous result for real quadratic fields was proved in the early 1970s independently by Yamamoto [18] and Weinberger [17]. Quantitative results giving a lower bound for the number of such fields were given later by Murty [15], Soundararajan [16], and Yu [19].

More generally, one can study the m-rank of the ideal class group. If A is a finitely generated abelian group, we define the m-rank of A, $\mathrm{rk}_m A$, to be the largest integer r

✉ Aaron Levin
levina@msu.edu

Kollin Bartz
bartzkol@msu.edu

Aman Dhruva Thamminana
thammina@msu.edu

[1] Department of Mathematics, Michigan State University, East Lansing, MI 48824, USA

such that A has a subgroup isomorphic to $(\mathbb{Z}/m\mathbb{Z})^r$. For a number field k, we let $\mathrm{Cl}(k)$ denote its ideal class group, and let d_k denote its (absolute) discriminant. Concentrating on the imaginary quadratic case, we let $\mathcal{N}^-(m^r; X)$ denote the number of imaginary quadratic number fields k satisfying $|d_k| \leq X$ and $\mathrm{rk}_m \mathrm{Cl}(k) \geq r$.

With this notation, improving on the results of Murty [15], Soundararajan [16] proved (for any $\epsilon > 0$),

$$\mathcal{N}^-(m; X) \gg \begin{cases} X^{\frac{1}{2}+\frac{2}{m}-\epsilon} & m \equiv 0 \pmod 4, \\ X^{\frac{1}{2}+\frac{3}{m+2}-\epsilon} & m \equiv 2 \pmod 4. \end{cases}$$

Since trivially $\mathcal{N}^-(m; X) \geq \mathcal{N}^-(2m; X)$, one also finds inequalities for odd m. For $m = 3$, better inequalities have been proved by Heath–Brown [8] and Lee et al. [11].

For rank $r = 2$, we have the following quantitative result of Kulkarni and the second author, slightly improving an earlier result of Byeon [3].

Theorem 1.1 (Kulkarni–Levin [10]) *Let $m > 1$ be an integer. Then*

$$\mathcal{N}^-(m^2; X) \gg X^{\frac{1}{m}}/(\log X)^2.$$

When m is odd, a better unconditional lower bound seems to be known only for $m = 3, 5, 7$ (when $m \geq 5$ is prime, Chattopadhyay and Saikia [6] have given a better lower bound conditional on the *abc*-conjecture). Improving on results of Luca and Pacelli [13] and Yu [19], Lee, Lee and Yoo [11] proved

$$\mathcal{N}^-(3^2; X) \gg X^{\frac{2}{3}}.$$

For $m = 5, 7$, Byeon proved

Theorem 1.2 (Byeon [4]) *We have*

$$\mathcal{N}^-(5^2; X) \gg X^{\frac{1}{4}},$$
$$\mathcal{N}^-(7^2; X) \gg X^{\frac{1}{4}}.$$

The goal of this note is the following improvement to Byeon's lower bound for $\mathcal{N}^-(5^2; X)$:

Theorem 1.3 *We have*

$$\mathcal{N}^-(5^2; X) \gg \frac{X^{\frac{1}{3}}}{(\log X)^2}.$$

Our method also yields an analogous lower bound for real quadratic fields whose class number is divisible by 5; however, we omit the details as this would not improve on the lower bound $\gg X^{\frac{1}{2}}$ in this setting due to Byeon [2].

The proof is based on the "geometric approach" to ideal class groups, which was formalized in work of the second author [12] and Gillibert and the second author [7]. More precisely, we use a recent enumerative result of Kulkarni and the second author (building on the work of [7]):

Theorem 1.4 (Kulkarni–Levin [10]) *Let C be a smooth projective hyperelliptic curve over $\mathbb{Q}$ with a $\mathbb{Q}$-rational Weierstrass point. Let $g = g(C)$ denote the genus of C and let $\mathrm{Jac}(C)(\mathbb{Q})_{\mathrm{tors}}$ denote the rational torsion subgroup of the Jacobian of C. Let $m > 1$ be an integer and let*

$$r = \mathrm{rk}_m \,\mathrm{Jac}(C)(\mathbb{Q})_{\mathrm{tors}}.$$

Then

$$\mathcal{N}^-(m^r; X) \gg \frac{X^{\frac{1}{g+1}}}{(\log X)^2}.$$

In view of Theorem 1.4, in order to prove Theorem 1.3 we only need to find a genus 2 curve C over $\mathbb{Q}$ with a $\mathbb{Q}$-rational Weierstrass point and

$$\mathrm{rk}_5 \,\mathrm{Jac}(C)(\mathbb{Q})_{\mathrm{tors}} \geq 2.$$

In fact, in this case we must have equality $\mathrm{rk}_5 \,\mathrm{Jac}(C)(\mathbb{Q})_{\mathrm{tors}} = 2$, since it is well known from a Weil pairing argument that if k does not contain an mth root of unity, then $\mathrm{rk}_m \,\mathrm{Jac}(C)(k)_{\mathrm{tors}} \leq g(C)$. To find a suitable genus 2 curve C we examine a construction of Howe, Leprévost, and Poonen [9], who studied the problem of constructing large torsion subgroups of Jacobians of curves of genus 2 and 3. Indeed, one of the aims of the "geometric approach" to ideal class groups is to take advantage of such constructions for rational torsion in Jacobians, and to leverage these arithmetic–geometric results to study ideal class groups of number fields.

2 A construction of Howe, Leprévost, and Poonen

We will construct a genus 2 curve C over $\mathbb{Q}$ with a rational Weierstrass point that admits two independent maps of degree 2 to elliptic curves E_1 and E_2, with each elliptic curve possessing a rational 5-torsion point. In this case, the maps $C \to E_1$ and $C \to E_2$ induce an isogeny between $\mathrm{Jac}(C)$ and $E_1 \times E_2$ of degree coprime to 5, and it follows that $\mathrm{rk}_5 \,\mathrm{Jac}(C)(\mathbb{Q})_{\mathrm{tors}} \geq 2$ (in fact, we have equality as noted above).

For this purpose, we use a construction of Howe, Leprévost, and Poonen [9] to produce many candidate curves C, with the exception that they may not possess a rational Weierstrass point. We then use a brute force search among the produced curves to find a curve C with a rational Weierstrass point.

We begin by summarizing the relevant facts. Recall that elliptic curves with a rational 10-torsion point form a 1-parameter family. Indeed, from [9, Table 6], the universal elliptic curve over $X_1(10)$ can be given explicitly as

$$E_t : y^2 = x(x^2 - (2t^2 - 2t + 1)(4t^4 - 12t^3 + 6t^2 + 2t - 1)x \\ +16(t^2 - 3t + 1)(t - 1)^5 t^5), \tag{2.1}$$

and from [9, Table 7] a point of order 10 on E_t is

$$(x, y) = (4(t - 1)(t^2 - 3t + 1)t^3, 4(t - 1)(t^2 - 3t + 1)t^3(2t - 1)).$$

The discriminant of E_t modulo squares is

$$\Delta_{10}(t) = (2t - 1)(4t^2 - 2t - 1).$$

Let E_t and E_u be two elliptic curves over $\mathbb{Q}$ with a rational 10-torsion point corresponding to the choices $t, u \in \mathbb{Q}$. From [9, Proposition 3], one can find a genus 2 curve C with Jacobian $\mathrm{Jac}(C)$ that is (2, 2)-isogenous to $E_t \times E_u$ if there is an isomorphism of Galois modules $E_t[2](\overline{\mathbb{Q}})$ to $E_u[2](\overline{\mathbb{Q}})$ that does not come from an isomorphism of elliptic curves (see [9, Proposition 3] for the precise statement). As noted in [9], since such an isomorphism must map the rational 2-torsion point to the rational 2-torsion point, the isomorphism exists if and only if the quadratic field defined by the non-rational 2-torsion points on E_t and E_u are equal, and this occurs if and only if the discriminants of E_t and E_u are equal modulo squares. Then the problem of finding suitable elliptic curves E_t and E_u reduces to finding rational solutions to the equation

$$\Delta_{10}(t)z^2 = \Delta_{10}(u). \tag{2.2}$$

In stating an explicit result, it will be slightly more convenient to use the equivalent equation

$$(2t - 1)^5(4t^2 - 2t - 1)z^2 = (2u - 1)^5(4u^2 - 2u - 1), \tag{2.3}$$

coming from considering the discriminant of the quadratic factor on the right-hand side of (2.1).

Then given a solution to (2.3), we have the following explicit result constructing a genus 2 curve C with Jacobian $\mathrm{Jac}(C)$ that is (2, 2)-isogenous to $E_t \times E_u$.

Theorem 2.1 *Let $(t, u, z) \in \mathbb{Q}^3$ be a solution to (2.3). Let*

$$\begin{aligned}
a &= 2(8t^6z - 32t^5z + 40t^4z - 20t^3z + 4tz - 8u^6 + 32u^5 - 40u^4 + 20u^3 - 4u - z + 1),\\
a_0 &= -64t^5(t-1)^5(t^2 - 3t + 1),\\
a_2 &= 2(32t^{12}z - 256t^{11}z + 832t^{10}z - 1440t^9z + 1440t^8z - 960t^7z + 64t^6u^6 - 256t^6u^5\\
&\quad + 320t^6u^4 - 160t^6u^3 + 32t^6u + 640t^6z - 8t^6 - 256t^5u^6 + 1024t^5u^5 - 1280t^5u^4\\
&\quad + 640t^5u^3 - 128t^5u - 480t^5z + 32t^5 + 320t^4u^6 - 1280t^4u^5 + 1600t^4u^4 - 800t^4u^3\\
&\quad + 160t^4u + 240t^4z - 40t^4 - 160t^3u^6 + 640t^3u^5 - 800t^3u^4 + 400t^3u^3 - 80t^3u - 40t^3z\\
&\quad + 20t^3 - 16t^2z + 32tu^6 - 128tu^5 + 160tu^4 - 80tu^3 + 16tu + 8tz - 4t - 8u^6 + 32u^5
\end{aligned}$$

$$
\begin{aligned}
&- 40u^4 + 20u^3 - 4u - z + 1),\\
a_4 &= 384t^7z^2 - 128t^6u^6z + 512t^6u^5z - 640t^6u^4z + 320t^6u^3z - 64t^6uz - 1152t^6z^2 + 16t^6z\\
&+ 512t^5u^6z - 2048t^5u^5z + 2560t^5u^4z - 1280t^5u^3z + 256t^5uz + 1344t^5z^2 - 64t^5z\\
&- 640t^4u^6z + 2560t^4u^5z - 3200t^4u^4z + 1600t^4u^3z - 320t^4uz - 720t^4z^2 + 80t^4z\\
&+ 320t^3u^6z - 1280t^3u^5z + 1600t^3u^4z - 800t^3u^3z + 160t^3uz + 120t^3z^2 - 40t^3z + 48t^2z^2\\
&- 64tu^6z + 256tu^5z - 320tu^4z + 160tu^3z - 32tuz - 24tz^2 + 8tz - 64u^{12} + 512u^{11}\\
&- 1664u^{10} + 2880u^9 - 2880u^8 + 1536u^7 + 16u^6z - 128u^6 - 64u^5z - 384u^5 + 80u^4z\\
&+ 240u^4 - 40u^3z - 40u^3 - 16u^2 + 8uz + 8u + 3z^2 - 2z - 1,\\
a_6 &= z(-128t^7z^2 + 384t^6z^2 - 448t^5z^2 + 240t^4z^2 - 40t^3z^2 - 16t^2z^2 + 8tz^2 + 64u^{12}\\
&- 512u^{11} + 1664u^{10} - 2880u^9 + 2880u^8 - 1536u^7 + 128u^6 + 384u^5 - 240u^4 + 40u^3\\
&+ 16u^2 - 8u - z^2 + 1).
\end{aligned}
$$

Suppose that $a \neq 0$ (or equivalently $z \neq \frac{8u^6-32u^5+40u^4-20u^3+4u-1}{8t^6-32t^5+40t^4-20t^3+4t-1}$) and

$$t, u \notin \left\{0, \frac{1}{2}, 1\right\}.$$

Then

$$y^2 = a(a_6x^6 + a_4x^4 + a_2x^2 + a_0)$$

defines a genus 2 curve C, $\mathrm{Jac}(C)$ *is $(2,2)$-isogenous to $E_t \times E_u$, and in particular,* $\mathrm{rk}_5 \operatorname{Jac}(C)(\mathbb{Q})_{\mathrm{tors}} = 2$. *Explicitly, E_t is isomorphic to the elliptic curve*

$$E_t' : y^2 = a(a_0x^3 + a_2x^2 + a_4x + a_6),$$

E_u is isomorphic to the elliptic curve

$$E_u' : y^2 = a(a_6x^3 + a_4x^2 + a_2x + a_0),$$

and the isogeny may be induced by the two bielliptic maps

$$
\begin{aligned}
C &\to E_t'\\
(x, y) &\mapsto (1/x^2, y/x^3)
\end{aligned}
$$

and

$$
\begin{aligned}
C &\to E_u'\\
(x, y) &\mapsto (x^2, y),
\end{aligned}
$$

respectively.

Theorem 2.1 is derived from Proposition 4 of [9]. However, once the computation is made, it is possible to give an independent direct (computational[1]) proof of the result, as we now describe.

Proof We first verify that $y^2 = a(a_6x^6 + a_4x^4 + a_2x^2 + a_0)$ defines a genus 2 curve C, or equivalently that $a(a_6x^6 + a_4x^4 + a_2x^2 + a_0)$ is a separable sextic polynomial. By assumption, $a \neq 0$. We have $a_6 = zf(t, u, z^2)$ for a certain polynomial f. Using (2.3) to eliminate z^2 in the polynomial f and factoring, we find that $a_6 \neq 0$ if $t, u \notin \left\{0, \frac{1}{2}, 1\right\}$ and neither t nor u are roots of $4x^2 - 2x - 1$ or $x^2 - 3x + 1$; in fact, this last condition is vacuous since $t, u \in \mathbb{Q}$ and the two quadratic polynomials are irreducible over $\mathbb{Q}$. Therefore $a(a_6x^6 + a_4x^4 + a_2x^2 + a_0)$ is a sextic polynomial. Finally, a direct computation gives that each irreducible factor (in $\mathbb{Q}[t, u, z]$) of the discriminant of $a_6x^6 + a_4x^4 + a_2x^2 + a_0$ divides one of $a, a_6, t, t-1, 2t-1, t^2-3t+1$, or $4t^2-2t-1$. Since none of these polynomials vanishes under our hypotheses, the sextic polynomial is separable.

It is well-known, going back to Jacobi in the nineteenth century, that $\mathrm{Jac}(C)$ is $(2, 2)$-isogenous to $E'_t \times E'_u$, induced by the given maps (see [5, Theorem 14.1.1]). Finally, by computing reduced Weierstrass forms of the curves, one can check by direct computation that E_t and $E_{t'}$ are isomorphic, and that E_u and E'_u are isomorphic (for the latter isomorphism, one must compute modulo the relation (2.3)). □

Remark 2.2 In fact, it follows from [9] that $\mathrm{Jac}(C)$ contains a rational torsion subgroup isomorphic to $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/10\mathbb{Z}$. One can see the extra 2-torsion factor directly from Theorem 2.1 as the polynomial $a(a_6x^6 + a_4x^4 + a_2x^2 + a_0)$ is easily checked to have the quadratic factor $zx^2 - 1$.

3 Search for a genus 2 curve with a rational Weierstrass point and $\mathrm{rk}_5\, \mathrm{Jac}(C)(\mathbb{Q})_{\mathrm{tors}} = 2$

We ran a naïve search for solutions to (2.3), and for each triple of solutions $(t, u, z) \in \mathbb{Q}^3$ satisfying the hypotheses of Theorem 2.1, we computed the hyperelliptic curve of Theorem 2.1 and checked if the curve C had a rational Weierstrass point. For t and u rational numbers of the form a/b with $|a|, |b| \leq 100$, we found[2] 21088 appropriate solutions $(t, u, z) \in \mathbb{Q}^3$, which led to 548 genus 2 curves with a rational Weierstrass point and $\mathrm{rk}_5\, \mathrm{Jac}(C)(\mathbb{Q}) = 2$, lying in 85 distinct isomorphism classes over $\mathbb{Q}$. For instance, the genus 2 curve corresponding to the solution

$$(t, u, z) = \left(\frac{2}{3}, -\frac{1}{3}, 25\right),$$

[1] Magma code verifying the calculations in the proof is available at https://github.com/amantham20/Rank5Quadratics-Research/blob/main/Magma_Code/Theorem21Computations.magma.

[2] Programs for searching for solutions and verifying the calculations of this section are available at https://github.com/amantham20/Rank5Quadratics-Research/tree/main and https://github.com/amantham20/Rank5Quadratics-Research/blob/main/Magma_Code/Section3Computations.magma.

may be given by the Weierstrass equation (after some simplification and changing to an odd model to make the rational Weierstrass point obvious):

$$\begin{aligned} C_0 : y^2 &= 640x^5 + 3641x^4 + 8878x^3 + 11729x^2 + 8392x + 2576 \\ &= (5x+7)(128x^4 + 549x^3 + 1007x^2 + 936x + 368). \end{aligned}$$

The computer algebra system Magma [1] verifies that the rational torsion subgroup of $\mathrm{Jac}(C_0)$ is exactly $\mathbb{Z}/5\mathbb{Z} \times \mathbb{Z}/10\mathbb{Z}$ (see Remark 2.2). Then the existence of the curve C_0 (or any of the other 84 found curves) combined with Theorem 1.4 finishes the proof of Theorem 1.3.

Author contributions All authors contributed equally to the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no conflict of interest.

References

1. Bosma, W., Cannon, J., Playoust, C.: The Magma algebra system. I. The user language. J. Symb. Comput. **24**(3–4), 235–265 (1997). (**Computational algebra and number theory (London, 1993)**)
2. Byeon, D.: Real quadratic fields with class number divisible by 5 or 7. Manuscr. Math. **120**(2), 211–215 (2006)
3. Byeon, D.: Imaginary quadratic fields with noncyclic ideal class groups. Ramanujan J. **11**(2), 159–163 (2006)
4. Byeon, D.: Quadratic fields with noncyclic 5- or 7-class groups. Ramanujan J. **19**(1), 71–77 (2009)
5. Cassels, J.W.S., Flynn, E.V.: Prolegomena to a Middlebrow Arithmetic of Curves of Genus 2. London Mathematical Society Lecture Note Series, vol. 230. Cambridge University Press, Cambridge (1996)
6. Chattopadhyay, J., Saikia, A.: On the p-ranks of the ideal class groups of imaginary quadratic fields. Ramanujan J. **62**(2), 571–581 (2023)
7. Gillibert, J., Levin, A.: Pulling back torsion line bundles to ideal classes. Math. Res. Lett. **19**(6), 1171–1184 (2012)
8. Heath-Brown, D.R.: Quadratic class numbers divisible by 3. Funct. Approx. Comment. Math. **37**(1), 203–211 (2007)
9. Howe, E.W., Leprévost, F., Poonen, B.: Large torsion subgroups of split Jacobians of curves of genus two or three. Forum Math. **12**(3), 315–364 (2000)
10. Kulkarni, K.R., Levin, A.: Hilbert's irreducibility theorem and ideal class groups of quadratic fields. Acta Arithm. **205**(4), 371–380 (2022)
11. Lee, S., Lee, Y., Yoo, J.: Infinite families of class groups of quadratic fields with 3-rank at least one: quantitative bounds. Int. J. Number Theory **19**(3), 621–637 (2023)

12. Levin, A.: Ideal class groups, Hilbert's irreducibility theorem, and integral points of bounded degree on curves. J. Théor. Nombres Bordeaux **19**(2), 485–499 (2007)
13. Luca, F., Allison, M.: Pacelli, Class groups of quadratic fields of 3-rank at least 2: effective bounds. J. Number Theory **128**(4), 796–804 (2008)
14. Nagel, T.: Über die Klassenzahl imaginär-quadratischer Zahlkörper. Abh. Math. Sem. Univ. Hamburg **1**(1), 140–150 (1922)
15. Ram Murty, M.: Exponents of class groups of quadratic fields. In: Topics in Number Theory (University Park, PA, 1997). Math. Appl., vol. 467, Kluwer, Dordrecht, pp. 229–239 (1999)
16. Soundararajan, K.: Divisibility of class numbers of imaginary quadratic fields. J. Lond. Math. Soc. **61**(3), 681–690 (2000)
17. Weinberger, P.J.: Real quadratic fields with class numbers divisible by n. J. Number Theory **5**, 237–241 (1973)
18. Yamamoto, Y.: On unramified Galois extensions of quadratic number fields. Osaka Math. J. **7**, 57–76 (1970)
19. Yu, G.: A note on the divisibility of class numbers of real quadratic fields. J. Number Theory **97**(1), 35–44 (2002)